

Roadmap - Topic 6 Weather in the Atmosphere

Learning Objectives	Vocabulary	Questions
Lesson 1: The Atmosphere Around You		
- the structure and layers of the Earth's atmosphere - the composition and characteristics of each layer in Earth's atmosphere - predict the stability and change caused by unequal heating of the Earth's atmosphere by the sun	Atmosphere Air pressure Troposphere Stratosphere	Altitude Wind Mesosphere Thermosphere 1. What gases make up the atmosphere? 2. How does altitude affect air pressure? 3. What are the defining characteristics of the layers of the atmosphere? 4. Model how convection currents occur and cause wind.
Lesson 2: Water in the Atmosphere		
- water vapor enters the atmosphere through a number of processes - water is continually evaporating and condensing in the atmosphere and this process forms clouds - precipitation is a vital part of the water cycle - water is always moving between the surface of Earth and the atmosphere - temperature determines the type of precipitation for an area	Water cycle Evaporation Condensation Precipitation Psychrometer Wet bulb Dry bulb	Dew point Humidity Relative humidity Snow Sleet Hail Freezing rain 1. How is dew point a measure of humidity? 2. What causes fog? 3. What is the difference between humidity and relative humidity? 4. How does temperature in the atmosphere affect precipitation?
Lesson 3: Air Masses		
- air masses are commonly moved by the prevailing winds and jet streams - air masses of different temperatures and humidity collide, resulting in a front - storms and changeable weather often develop along fronts - the type of front that develops depends on the characteristics of the air masses and the direction in which they move	Air mass Jet stream Front Prevailing Stationary	Cyclone Anticyclone Continental Maritime Polar Tropical 1. What is the general direction that air masses move across the continental US? 2. What are the differences between continental and maritime air masses? 3. What are fronts and what do they all have in common? 4. What types of weather are low pressure systems associated with? High pressure?

Lesson 4: Predicting Weather Changes

- use observations, patterns, and tools to predict the weather
- determine the effect of global patterns on local weather by observing and tracking the movements of jet streams and ocean currents

Meteorologist

1. How does the Gulf Stream affect temperatures on the East Coast?
2. Why is predicting weather complicated?